

SmartDesk AI

Test Cases & Functional Verification Report

Version 1.0 | June 2026

Test Cases

Test Case ID	Input Scenario	Expected Category	Expected Priority	Result
TC001	Audio upload triggers transcription pipeline	Audio Ingestion	P1	Pass
TC002	Groq Whisper returns transcription text	Transcription	P1	Pass
TC003	Mock mode runs offline with no API key	Mock Mode	P2	Pass
TC004	LLaMA extracts correct ticket fields from transcript	AI Extraction	P1	Pass
TC005	Ticket saved to SQLite database on submit	Database	P1	Pass
TC006	Duplicate ticket detection flags similar issues	Validation	P2	Pass
TC007	Priority assigned correctly (Critical / High / Medium / Low)	AI Extraction	P1	Pass
TC008	Department routing email sent via SMTP	Email Service	P2	Pass
TC009	Ticket status updates from Pending to WIP to Resolved	Dashboard	P1	Pass
TC010	CSV export downloads all filtered tickets	Dashboard	P3	Pass

Functional Testing

Audio Ingestion & Transcription

- Verify .mp3, .wav, and .m4a files are accepted by the uploader.
- Verify transcription output is returned as readable text.
- Verify Mock Mode returns expected preset transcript by filename keyword.
- Verify error message is shown for unsupported file formats.

AI Ticket Extraction

- Verify all 12 structured fields are extracted (name, ID, email, priority, etc.).
- Verify correct category is assigned from the 8 supported departments.
- Verify sentiment field is populated (Positive / Neutral / Negative).
- Verify LLaMA output is valid JSON and parsed without crash.

Database & CRUD

- Verify tickets.db is auto-created on first launch.
- Verify new tickets are inserted with unique SMD-prefixed IDs.

- Verify status update persists after page reload.
- Verify ticket assignment to engineer is saved correctly.

Dashboard & UI

- Verify Home page KPI cards show correct ticket counts.
- Verify Database page table renders all tickets with filters.
- Verify search bar filters tickets by title and description.
- Verify status dropdown updates reflect immediately in the table.

Email Service

- Verify routing email is dispatched on new ticket creation.
- Verify correct department email address is used per category.
- Verify SMTP failure is caught and logged without crashing the app.

Performance Testing

- Process 20 audio tickets consecutively within 60 seconds.
- Database insertion success rate: 100% across 100 records.
- Mock Mode ticket generation completes in under 2 seconds per ticket.
- Dashboard loads with 500+ tickets without UI lag or crash.

Error Handling Testing

- Empty audio file uploaded — show appropriate error message.
- Invalid GROQ_API_KEY — graceful fallback to Mock Mode or error alert.
- LLaMA returns malformed JSON — rule-based fallback activates.
- Database file locked / corrupted — error is caught and logged.
- SMTP credentials missing — email skipped, ticket still created.
- Missing employee fields in transcript — defaults applied, ticket saved.